

# CSS Border-Radius

Complete Guide: Creating Rounded Corners and Complex Shapes

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# 1 Introduction

The CSS `border-radius` property creates rounded corners on elements. It's one of the most widely used CSS properties for modern web design, enabling everything from subtle rounded corners to complex circular and elliptical shapes.

## 1.1 Historical Context

Before `border-radius` became standard (CSS3), creating rounded corners required:

- Multiple nested divs with background images
- Complex image slicing techniques
- Heavy reliance on graphics software
- Limited flexibility and maintainability

Today, `border-radius` provides an elegant, scalable solution for rounded corners across all modern browsers.

## 1.2 Common Use Cases

- Card designs with rounded corners
- Circular profile pictures
- Button styling
- Image galleries
- Modal dialogs
- Tab interfaces
- Avatar thumbnails
- Modern UI elements

## 2 Basic Border-Radius

### 2.1 Simple Rounded Corners

The most basic usage creates uniform rounded corners on all four corners:

```
.box {  
  width: 200px;  
  height: 200px;  
  background-color: lightblue;  
  border-radius: 10px; /* 10px rounded corners on all sides */  
  border: 2px solid navy;  
}
```

### 2.2 Percentage Values

Border-radius can use percentage values, which are relative to the element's dimensions:

```
.box {  
  width: 200px;  
  height: 100px;  
  border-radius: 50%; /* Creates an ellipse */  
}  
  
.square {  
  width: 200px;  
  height: 200px;  
  border-radius: 50%; /* Creates a perfect circle */  
}
```

### 2.3 Unit Types

| Unit           | Example          | Use Case                   |
|----------------|------------------|----------------------------|
| Pixels (px)    | 10px, 15px, 25px | Fixed size corners         |
| Percentage (%) | 10%, 25%, 50%    | Relative to dimensions     |
| EM             | 0.5em, 1em       | Relative to font size      |
| REM            | 0.5rem, 1rem     | Relative to root font size |

## 3 Individual Corner Control

### 3.1 Four Corners Separately

Control each corner individually using specific properties:

#### 3.1.1 The Four Properties

- `border-top-left-radius`
- `border-top-right-radius`
- `border-bottom-right-radius`
- `border-bottom-left-radius`

### 3.1.2 Example: Individual Corners

```
.custom-corner-box {  
  width: 300px;  
  height: 200px;  
  border-top-left-radius: 20px;  
  border-top-right-radius: 0px;  
  border-bottom-right-radius: 30px;  
  border-bottom-left-radius: 10px;  
  border: 2px solid blue;  
}
```

## 3.2 Shorthand Syntax

The `border-radius` shorthand follows a specific order:

```
/* 1 value: All corners */  
border-radius: 10px;  
  
/* 2 values: Top-left & bottom-right | Top-right & bottom-left */  
border-radius: 10px 20px;  
  
/* 3 values: Top-left | Top-right & bottom-left | Bottom-right */  
border-radius: 10px 20px 30px;  
  
/* 4 values: Top-left | Top-right | Bottom-right | Bottom-left */  
border-radius: 10px 20px 30px 40px;
```

## 3.3 Examples

```
/* Uniform rounded corners */  
.card {  
  border-radius: 8px;  
}  
  
/* Top rounded, bottom square */  
.top-rounded {  
  border-radius: 8px 8px 0px 0px;  
}  
  
/* Different radius for each corner */  
.asymmetric {  
  border-radius: 5px 15px 25px 35px;  
}  
  
/* Only top-left rounded */  
.one-corner {  
  border-radius: 15px 0px 0px 0px;  
}
```

## 4 Elliptical Border-Radius

### 4.1 Horizontal and Vertical Radii

Border-radius can define both horizontal and vertical curves independently using the slash syntax:

```
border-radius: horizontal / vertical;
```

### 4.2 Ellipse Examples

```
/* Horizontal ellipse */
.wide-ellipse {
  width: 300px;
  height: 100px;
  border-radius: 50% / 25%;
}

/* Tall ellipse */
.tall-ellipse {
  width: 100px;
  height: 300px;
  border-radius: 25% / 50%;
}

/* Different curves for each corner */
.complex-ellipse {
  border-radius:
    40% 60% / 40% 60%;
}

/* Four independent ellipses */
.four-ellipses {
  width: 200px;
  height: 300px;
  border-radius:
    20px 40px 60px 80px /
    30px 50px 10px 90px;
}
```

### 4.3 Mathematical Understanding

The slash separator creates independent horizontal and vertical curves:

```
/* Before slash: horizontal radius
   After slash: vertical radius */

border-radius: 50px 100px / 100px 50px;

/* Expands to:
   Horizontal: top-left=50px, top-right=100px,
               bottom-right=50px, bottom-left=100px
   Vertical:   top-left=100px, top-right=50px,
               bottom-right=100px, bottom-left=50px
```

```
*/
```

## 5 Real-World Examples

### 5.1 Example 1: Card Design

```
.card {  
  width: 300px;  
  background-color: white;  
  border-radius: 8px;  
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);  
  overflow: hidden;  
}  
  
.card-image {  
  width: 100%;  
  height: 200px;  
  object-fit: cover;  
}  
  
.card-content {  
  padding: 20px;  
}  
  
/* Creates a modern card with rounded corners */
```

### 5.2 Example 2: Perfect Circle Avatar

```
.avatar {  
  width: 100px;  
  height: 100px;  
  border-radius: 50%; /* Makes it circular */  
  border: 3px solid #3498db;  
  object-fit: cover; /* For images */  
}  
  
.avatar-large {  
  width: 150px;  
  height: 150px;  
  border-radius: 50%;  
  background-color: lightgray;  
}
```

### 5.3 Example 3: Rounded Button

```
.button {  
  padding: 12px 24px;  
  background-color: #3498db;  
  color: white;
```

```
border: none;
border-radius: 5px;
cursor: pointer;
font-size: 16px;
transition: background-color 0.3s;
}

.button:hover {
  background-color: #2980b9;
}

.button-pill {
  border-radius: 25px; /* Pill-shaped button */
}
```

## 5.4 Example 4: Image Gallery

```
.gallery-container {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 15px;
}

.gallery-item {
  width: 100%;
  height: 250px;
  border-radius: 10px;
  overflow: hidden;
  cursor: pointer;
  transition: transform 0.3s;
}

.gallery-item img {
  width: 100%;
  height: 100%;
  object-fit: cover;
}

.gallery-item:hover {
  transform: scale(1.05);
}
```

## 5.5 Example 5: Tab Interface

```
.tabs {
  display: flex;
  gap: 0;
}

.tab {
  padding: 12px 24px;
```



```
background-color: #ecf0f1;
border: none;
cursor: pointer;
border-top-left-radius: 8px;
border-top-right-radius: 8px;
}

.tab.active {
background-color: white;
border-bottom: 3px solid #3498db;
}

.tab-content {
background-color: white;
padding: 20px;
border-bottom-left-radius: 8px;
border-bottom-right-radius: 8px;
}
```

## 6 Border-Radius with Other CSS Properties

### 6.1 Combining with Border

```
.box-with-border {  
  width: 200px;  
  height: 200px;  
  border: 3px solid #3498db;  
  border-radius: 15px;  
  padding: 20px;  
}  
  
/* Border follows the rounded corners */
```

### 6.2 Combining with Box-Shadow

```
.shadowed-box {  
  width: 300px;  
  border-radius: 10px;  
  background-color: white;  
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.15);  
}  
  
/* Shadow respects the rounded shape */
```

### 6.3 Combining with Background Gradient

```
.gradient-box {  
  width: 300px;  
  height: 200px;  
  border-radius: 15px;  
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);  
  padding: 20px;  
  color: white;  
}  
  
/* Gradient respects rounded corners */
```

### 6.4 Combining with Images

```
.rounded-image {  
  width: 300px;  
  height: 200px;  
  border-radius: 10px;  
  overflow: hidden; /* Important! */  
}  
  
.rounded-image img {  
  width: 100%;  
  height: 100%;
```

```
    object-fit: cover;
}

/* Image inside respects the border-radius */
```

## 6.5 Combining with Transform

```
.animated-box {
  width: 200px;
  height: 200px;
  border-radius: 10px;
  background-color: lightblue;
  transition: all 0.3s ease;
}

.animated-box:hover {
  transform: rotate(5deg) scale(1.1);
  border-radius: 20px; /* More rounded on hover */
}
```

# 7 Creating Complex Shapes

## 7.1 Circles and Ellipses

```
/* Perfect circle */
.circle {
  width: 200px;
  height: 200px;
  border-radius: 50%;
}

/* Horizontal ellipse */
.ellipse-h {
  width: 300px;
  height: 150px;
  border-radius: 50%;
}

/* Vertical ellipse */
.ellipse-v {
  width: 100px;
  height: 300px;
  border-radius: 50%;
}
```

## 7.2 Leaf Shape

```
.leaf {
  width: 200px;
```

```
height: 200px;
border-radius: 0 100%;
background-color: green;
}
```

### 7.3 Teardrop Shape

```
.teardrop {
width: 100px;
height: 150px;
border-radius: 50% 50% 50% 0;
transform: rotate(-45deg);
background-color: blue;
}
```

### 7.4 Petal Shape

```
.petal {
width: 100px;
height: 100px;
border-radius: 50% 0;
background-color: pink;
}
```

## 8 Browser Support & Compatibility

### 8.1 Standard Properties

| Property       | Chrome | Firefox | Safari | Edge | IE    |
|----------------|--------|---------|--------|------|-------|
| border-radius  | All    | 3+      | 5+     | All  | IE 9+ |
| Elliptical (/) | All    | 3.5+    | 5+     | All  | IE 9+ |

### 8.2 Vendor Prefixes

Modern browsers support border-radius without prefixes. For older browser support:

```
.box {
-webkit-border-radius: 10px; /* Safari 4, iOS 3.2 */
-moz-border-radius: 10px; /* Firefox 3.6 */
border-radius: 10px; /* Standard */
}
```

## 9 Performance Considerations

### 9.1 Rendering Performance

- Border-radius has minimal performance impact
- Hardware acceleration works well with rounded corners

- Combining with box-shadow may require more computation
- Modern browsers optimize rounded corner rendering

## 9.2 Best Practices

1. Use `overflow: hidden` for images inside rounded containers
2. Avoid excessive nesting with rounded corners
3. Use consistent border-radius values across designs
4. Test complex shapes on different browsers
5. Consider mobile rendering performance

## 10 Advanced Techniques

### 10.1 Asymmetric Corners

Create interesting asymmetric designs:

```
.asymmetric {
  width: 300px;
  height: 200px;
  border-radius: 50px 0 30px 10px;
  background: linear-gradient(45deg, #667eea, #764ba2);
}
```

### 10.2 Responsive Border-Radius

Scale border-radius with viewport:

```
.responsive-box {
  width: 100%;
  max-width: 400px;
  border-radius: 2vw; /* Scale with viewport width */
}

@media (max-width: 768px) {
  .responsive-box {
    border-radius: 5vw; /* Larger on mobile */
  }
}
```

### 10.3 Border-Radius Animation

Animate between different border-radius values:

```
.morphing-box {
  width: 200px;
  height: 200px;
  background-color: #3498db;
  border-radius: 10px;
  transition: border-radius 0.5s ease;
}

.morphing-box:hover {
  border-radius: 50%; /* Becomes circular */
}

.morphing-box:active {
  border-radius: 0; /* Back to square */
}
```

### 10.4 Clip-Path Alternative

For more complex shapes, consider clip-path:

```
/* Border-radius: Simple, smooth curves */
.rounded {
  border-radius: 20px;
}

/* Clip-path: More complex, polygonal shapes */
.complex-shape {
  clip-path: polygon(50% 0%, 100% 25%, 100% 75%,
                    50% 100%, 0% 75%, 0% 25%);
}
```

## 11 Common Mistakes & Solutions

### 11.1 Mistake 1: Forgetting Overflow Hidden

**Bad:**

```
.image-box {
  border-radius: 10px;
  background-color: lightgray;
}

.image-box img {
  width: 100%;
  /* Image corners not rounded - appears square */
}
```

**Good:**

```
.image-box {
  border-radius: 10px;
  overflow: hidden; /* Clips image to rounded corners */
}

.image-box img {
  width: 100%;
}
```

### 11.2 Mistake 2: Circular Size Mismatch

**Bad:**

```
.circle {
  width: 200px;
  height: 150px; /* Different from width */
  border-radius: 50%; /* Creates ellipse, not circle */
}
```

**Good:**

```
.circle {
  width: 200px;
  height: 200px; /* Same as width */
}
```

```
border-radius: 50%; /* Perfect circle */
}
```

### 11.3 Mistake 3: Wrong Unit in Shorthand

**Bad:**

```
.box {
  border-radius: 10px / 20 / 5px; /* Confusing mixing */
}
```

**Good:**

```
.box {
  border-radius: 10px / 20px; /* Clear and consistent */
}
```

## 12 Integration with Modern Frameworks

### 12.1 Tailwind CSS

Tailwind provides utility classes for border-radius:

```
<!-- Rounded corners -->
<div class="rounded">...</div>

<!-- Specific corners -->
<div class="rounded-lg">...</div>

<!-- Full circle -->
<div class="rounded-full">...</div>

<!-- Different sides -->
<div class="rounded-t-lg">...</div>
```

### 12.2 Bootstrap

Bootstrap includes border-radius utilities:

```
<!-- Default rounded corners -->
<div class="rounded">...</div>

<!-- Large rounded corners -->
<div class="rounded-lg">...</div>

<!-- Pill shape -->
<div class="rounded-pill">...</div>

<!-- Circle -->
<div class="rounded-circle">...</div>
```



## 13 Responsive Border-Radius Patterns

### 13.1 Mobile vs Desktop

```
.card {  
  border-radius: 8px;  
  width: 100%;  
}  
  
@media (max-width: 768px) {  
  .card {  
    border-radius: 12px; /* Larger on mobile */  
  }  
}  
  
@media (min-width: 1200px) {  
  .card {  
    border-radius: 6px; /* Smaller on desktop */  
  }  
}
```

## 14 Accessibility Considerations

### 14.1 Visual Clarity

- Ensure sufficient contrast between rounded elements and background
- Avoid excessive border-radius that obscures content
- Use consistent radius values for user recognition
- Test with various color schemes and themes

### 14.2 Focus Indicators

```
.button {  
  border-radius: 8px;  
  outline: none;  
  transition: box-shadow 0.3s;  
}  
  
.button:focus {  
  box-shadow: 0 0 0 3px rgba(52, 152, 219, 0.5);  
}  
  
/* Focus indicator respects rounded corners */
```

## 15 Summary: Key Takeaways

### 15.1 Border-Radius Values

| Value    | Effect             | Use Case         |
|----------|--------------------|------------------|
| 0px      | Square (default)   | Sharp corners    |
| 5px-15px | Subtle rounding    | Cards, buttons   |
| 25px+    | Prominent rounding | Avatars, pills   |
| 50%      | Circle/ellipse     | Avatars, bubbles |

### 15.2 Best Practices

1. Use `overflow: hidden` for images in rounded containers
2. Keep equal width/height for perfect circles (use 50%)
3. Maintain consistency with border-radius values
4. Test on different browsers and devices
5. Use percentage values (50%) for responsive circular shapes
6. Combine with transitions for smooth animations
7. Remember the shorthand order: top-left, top-right, bottom-right, bottom-left
8. Use the slash syntax for elliptical shapes

### 15.3 Quick Reference

```
/* Uniform rounded corners */
border-radius: 10px;

/* Different corners */
border-radius: 10px 20px 30px 40px;

/* Perfect circle */
border-radius: 50%;

/* Ellipse with aspect ratio */
border-radius: 50% / 25%;

/* Individual corners */
border-top-left-radius: 10px;
```

## 16 Conclusion

The CSS `border-radius` property is a fundamental tool for modern web design. From subtle rounded corners on cards to bold circular avatars, border-radius enables elegant, polished interfaces with minimal code.

Key principles to remember:

1. Simple rounded corners improve user experience
2. Use 50% for perfect circles
3. Remember to use `overflow: hidden` with images
4. Combine with other properties for advanced effects
5. Test responsive designs across devices
6. Maintain consistent values for visual cohesion

With mastery of border-radius, you'll create modern, professional-looking web interfaces that users love.

Happy coding!